

JAEEK SHIN

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🔍 google scholar

🌐 personal website

🌐 linkedin

EDUCATION

B.S., Mathematics, Seoul National University	2014-2018
M.S., Mathematics, Seoul National University	2018-2020
Ph.D., Electrical and Computer Engineering, Seoul National University	2020-Present

PUBLICATIONS

* indicates equal contribution.

Journal Publications

- Hamilton-Jacobi deep Q-learning for deterministic continuous-time systems with Lipschitz continuous controls
Jeongho Kim, **Jaek Shin**, and Insoon Yang
Journal of Machine Learning Research (JMLR), vol. 22, no. 206, pp. 1-34, 2021.
- Infusing model predictive control into meta-reinforcement learning for mobile robots in dynamic environments
Jaek Shin, Astghik Hakobyan, Mingyu Park, Yeoneung Kim, Gihun Kim, and Insoon Yang
IEEE Robotics and Automation Letters (RA-L), vol. 7, issue 4, pp. 10065-10072, 2022. (Selected for presentation at *IROS 22*)
- Control of Fab Lifters via Deep Reinforcement Learning: A Semi-MDP Approach
Giho Kim*, **Jaek Shin***, Gihun Kim, Joonrak Kim, and Insoon Yang
IEEE Transactions on Automation Science and Engineering (T-ASE), vol. 21, issue 4, pp. 5136-5148, 2024.
- Anderson acceleration for partially observable Markov decision processes: A maximum entropy approach
Mingyu Park*, **Jaek Shin***, and Insoon Yang
Automatica, vol. 164, 2024.
- Multiparametric Analysis of Multi-Task Markov Decision Processes: Structure, Invariance, and Reducibility
Jaek Shin and Insoon Yang
IEEE Control System Letters (L-CSS), vol. 8, pp. 928-933, 2024. (Selected for presentation at *CDC 24*)

Conference Publications

- On task-relevant loss functions in meta-reinforcement learning
Jaek Shin, Giho Kim, Howon Lee, Joonho Han, and Insoon Yang
Annual Learning for Dynamics & Control Conference (L4DC), 2024.
- Egocentric Conformal Prediction for Safe and Efficient Navigation in Dynamic Cluttered Environments
Jaek Shin, Jungjin Lee, and Insoon Yang
IEEE Conference on Decision and Control (CDC), 2025. (accepted)

ACADEMIC EXPERIENCE

Graduate Teaching Assistant

- Undergraduate Calculus II *Fall 2018, SNU*
- Linear Algebra *Spring 2019, SNU*
- Mathematical Theory and Algorithms for Optimization *Fall 2019, SNU*
- Advanced Control Techniques *Fall 2020, SNU*
- Stochastic Control and Reinforcement Learning *Spring 2021, SNU*
- Stochastic Control and Reinforcement Learning *Spring 2021, SNU*
- Advanced Control Techniques *Fall 2022, SNU*
- Linear Algebra *Fall 2024, SNU*

Research Experience

December 2020 - September 2021

SK hynix

Seoul, South Korea

- Design of efficient scheduling algorithm of FAB lifters using deep reinforcement learning

Teaching Assistant

August 2020, July 2021

Samsung Electronics

Seoul, South Korea

- Deep reinforcement learning lecture in Samsung AI Expert Educational Program

ACADEMIC SERVICES

Reviewer for Journals

- *IEEE Transactions on Robotics (T-RO)*
- *IEEE Transactions on Automation Science and Engineering (T-ASE)*
- *IEEE Robotics and Automation Letters (RA-L)*
- *IEEE Control System Letters (L-CSS)*

Reviewer for Conferences

- *International Conference on Learning Representations (ICLR)*
- *ACM International Conference on Hybrid Systems: Computation and Control (HSCC)*
- *American Control Conference (ACC)*
- *IEEE Conference on Decision and Control (CDC)*
- *Annual Learning for Dynamics and Control Conference (L4DC)*
- *Conference on Robot Learning (CoRL)*

TECHNICAL STRENGTHS

Computer Languages

Python, MATLAB, C++